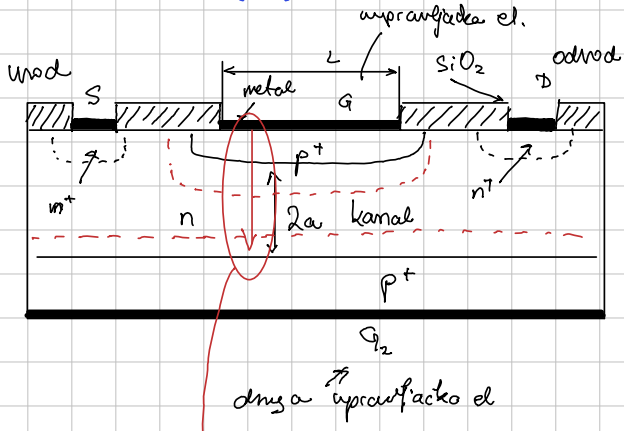
Struktura spojnog FET-a

n kanalni JFET



povećanje U_{reverz} → širina osiromaš. područja se povećava

→ unutar osiromaš. područja nema nosilaca

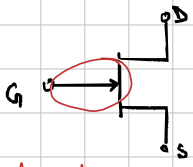
↳ sa širinom osiromaš. područja kontroliramo otvorenost/zatvorenost kanala

⇒ vodljivost

→ kompatibilno sa BipT → često idu zajedno

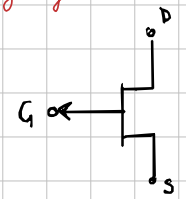
El. simboli JFET-a

strelica uvijek gleda p → n



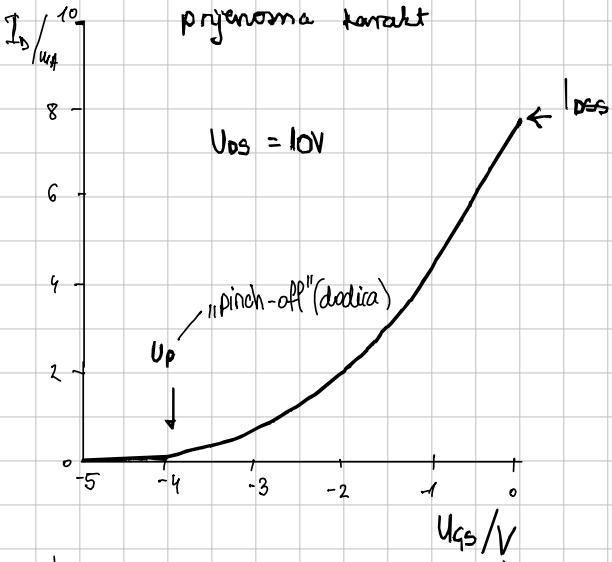
od p do n

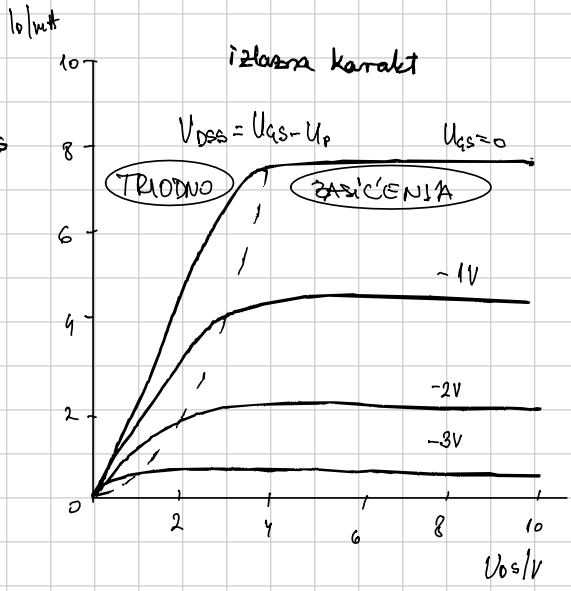
n-kanalni



p-kanalni

Karakteristike JFET-a



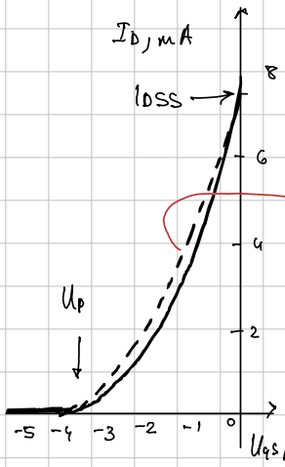


I_{DSS} → max struja JFET-a → za $U_{GS}=0$

U_P → napon dodira

↳ dolazi do prekida kanala

JFET u području zasićenja



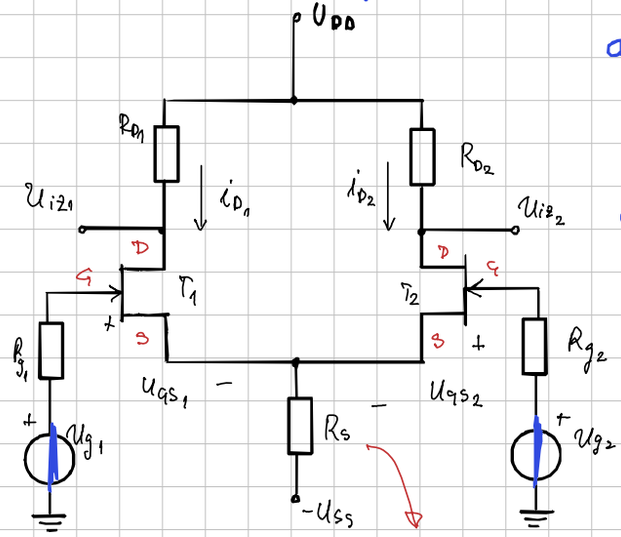
→ u pojačalima - radi u području zasićenja

aproximacija (pojednostavljeni izraz)

$$\boxed{I_D = I_{DS} = I_{DSS}\left(1 - \frac{U_G}{U_P}\right)^2}$$

Unipolarno dif. pojačalo s uvodskim otpornikom

→ spoj zajedničkog vrela



asimetrican ulaz

↳ samo u_{iz_1} ili u_{iz_2}

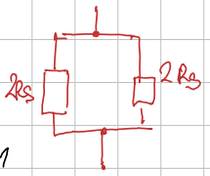
diferencijski ili simetrični ulaz

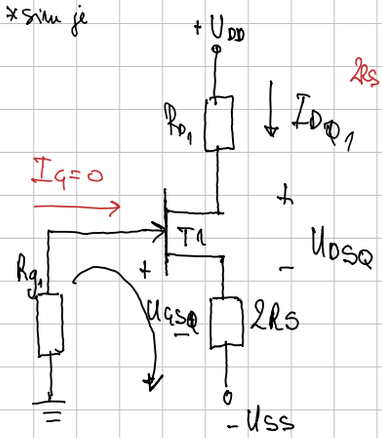
↳ razlika $u_{iz} = u_{iz_2} - u_{iz_1}$

Primjer 1.7) → Statika

K.S. AC

*sim je





$U_P = -4V$ $\quad R_S = R_{D_1} = R_{D_2} = 5k\Omega$

$I_{DSS} = 8mA$ $\quad R_{g_1} = R_{g_2} = 1k\Omega$

1) $U_{GSQ} + I_{DQ} \cdot 2R_S - U_{SS} = 0$

2) $-U_{DD} + I_{D_1}(R_{D_1} + 2R_S) + U_{DSQ} - U_{SS} = 0$

3) * $I_{DQ} = I_{SS}\left(1 - \frac{U_{GSQ}}{U_P}\right)^2$ $\quad$ * $I_{DSS} = I_{SS}$

$$U_{GSQ} = U_{SS} - I_{DQ} \cdot 2R_S = U_{SS} - I_{SS}\left(1 - \frac{U_{GSQ}}{U_P}\right)^2 \cdot 2R_S$$! riješavati po U_{GSQ}!

$$U_{GSQ} = U_{SS} - I_{SS}\left(1 - 2\frac{U_{GSQ}}{U_P} + \frac{U_{GSQ}^2}{U_P^2}\right) \cdot 2R_S$$

$$U_{GSQ} = 18V - 8\times10^{-3}\left(1 + 2\cdot\frac{U_{GSQ}}{4} + \frac{U_{GSQ}^2}{16}\right)\cdot 2\cdot 5\cdot 1000$$

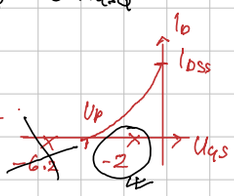
$$U_{GSQ} = 18 - 80\left(1 + \frac{U_{GSQ}}{2} + \frac{U_{GSQ}^2}{16}\right) = 18 - 80 - 40U_{GSQ} - 5U_{GSQ}^2$$

$$5U_{GSQ}^2 + 41U_{GSQ} + 62 = 0$$

↳ $U_{GSQ1} = -6.2V$

$U_{GSQ2} = -2V$ → kanal osiromašen

koji je prihvatljiv? n-kanal JFET:



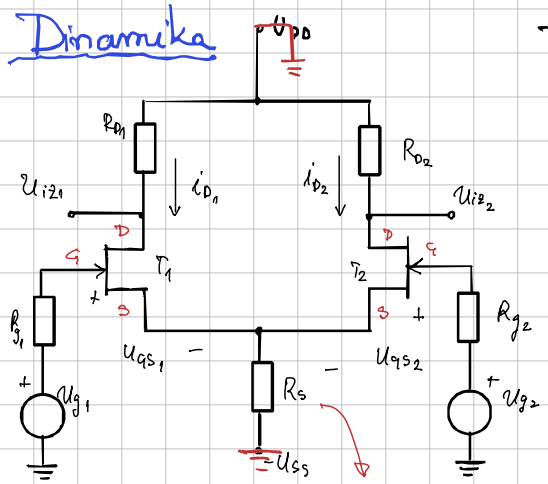
$$I_{DQ} = \frac{U_{SS} - U_{GSQ}}{2R_S} = \dots = 2mA$$

$$U_{DSQ} = U_{DD} - (-U_{SS}) - I_{DQ}\cdot(R_D + 2R_S) = \dots = 6V$$

$6V = |U_{DSQ}|$ $\quad |U_{GSQ} - U_P| = |-2 - (-4)| = 2V$

$6V > 2V$ ⇒ područje zasićenja

Dinamika



Dinamički parametri

$$I_D = I_{SS}\left(1-\frac{U_{GS}}{U_P}\right)^2(1+\lambda U_{DS})$$

faktor modulacije duljine kanala

$$g_m = \left.\frac{\partial i_D}{\partial u_{GS}}\right|_Q = 2\frac{I_{SS}}{-U_P}\left(1-\frac{U_{GS}}{U_P}\right)(1+\lambda U_{DS})$$

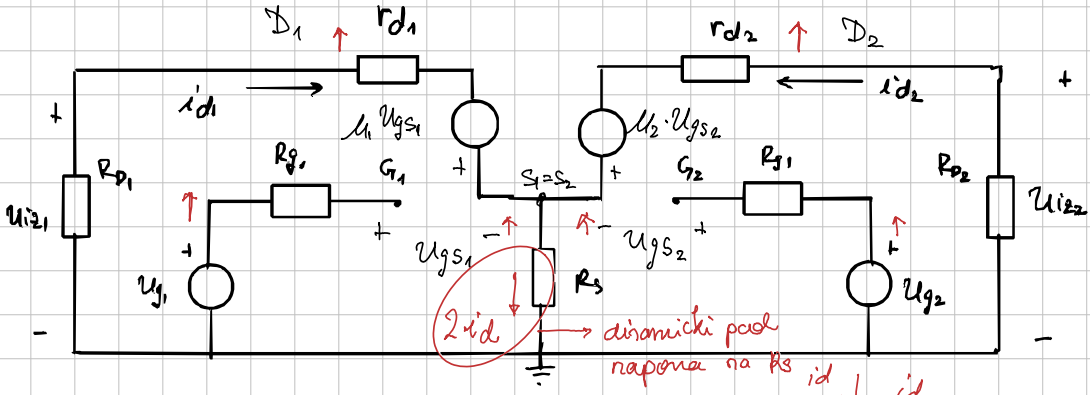
$$= 2\frac{I_{SS}}{-U_P}\left(1-\frac{U_{GSQ}}{U_P}\right)(1+\lambda U_{DSQ})$$

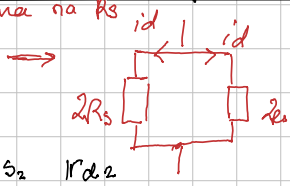
$g_m = 2{,}02\ mA/V$

$r_d = 250\ k\Omega$

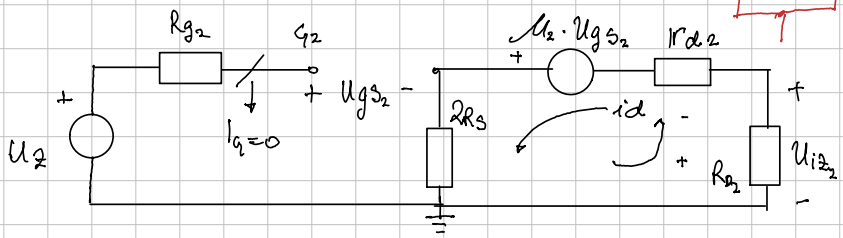
$\mu = 505$

$$r_d = \frac{1}{\left.\frac{\partial i_D}{\partial u_{DS}}\right|_Q} = \frac{1}{\lambda\cdot \underbrace{I_{SS}\left(1-\frac{U_{GSQ}}{U_P}\right)^2}_{I_{DQ}}} = \frac{1}{\lambda I_{DQ}}$$





ZAJEDNIČKI SIGNAL:



$$A_{vz_2} = \frac{U_{iz_2}}{U_z} \qquad U_{iz_2} = -i_d\cdot R_{D_2} \qquad -\mu_2 U_{gs_2} + i_d\,(2R_S + R_2 + r_{d_2}) = 0$$

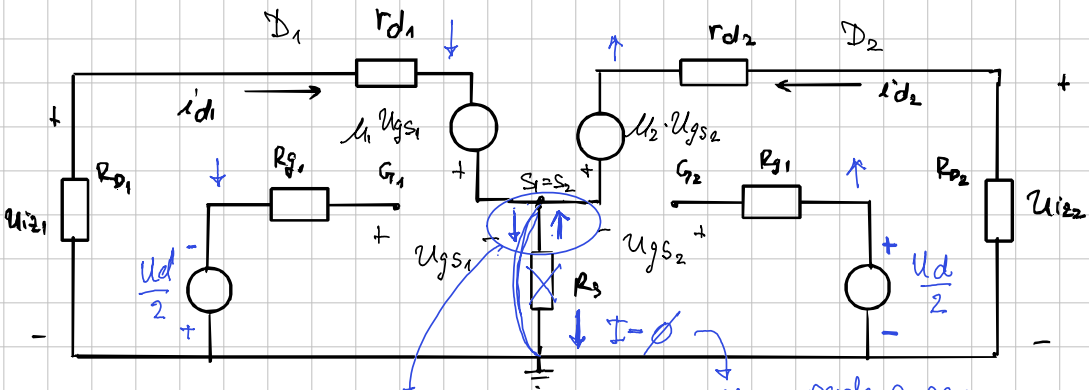
$$-U_z + \cancel{i_d R_{g_2}} + U_{gs_2} + i_d\cdot 2R_S = 0 \longrightarrow U_z = U_{gs} + i_d\cdot 2R_S \rightarrow U_{gs} = U_z - i_d\cdot 2R_S$$

$$i_d(2R_S + R_{D_2} + r_{d_2}) = \mu\cdot U_z - i_d\cdot 2R_S\cdot\mu \Rightarrow i_d\,(2R_S(1+\mu) + R_{D_2} + r_{d_2}) = \mu U_z$$

$$\Rightarrow A_{vz} = \frac{-i_d\cdot R_{D_2}\cdot\mu}{i_d\cdot(2R_S(1+\mu) + R_{D_2} + r_{d_2})} = \boxed{-\mu\frac{R_{D_2}}{2R_S(1+\mu) + R_{D_2} + r_{d_2}} = A_{vz}} = \dots = -0.475$$

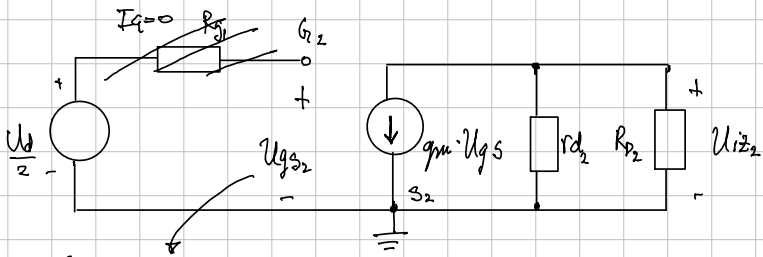
degeneracija

DIFERENCIJSKI SIGNALI



jednake amplitude ali različite faze

nema pada napona na RS



$$A_{vd} = \frac{u_{iz_2}}{u_d} = \frac{1}{2} \cdot \frac{u_{iz_2}}{\frac{u_d}{2}}$$

$$u_{gs_2} = \frac{u_d}{2}$$

$$u_{iz_2} = -g_m \cdot u_{gs} (r_{d2} \parallel R_{D2}) \quad , \quad \frac{u_d}{2} = u_{gs}$$

$$\boxed{A_{vd_2} = -\frac{1}{2} g_m (r_{d2} \parallel R_{D2})} = \dots = -4.95$$

$$\rho = \left| \frac{A_{vd}}{A_{vz}} \right| = \left| \frac{-4.95}{-0.475} \right| = \dots = 10.4 = 20.4 \text{ dB}$$

puno manji

↳ strmine unipol. su puno manje od lipsića

$$\rho = \dots = (1+\mu_2) \frac{R_S}{r_{d2} + R_{D2}} + \frac{1}{2}$$

izlazni napon → zbroj napona uz diferencijski i zajednički signal